

WiFi Training Program

Assignment Report – Module 1

Q1. In which OSI layer does the Wi-Fi standard/protocol fit?

Wi-Fi (IEEE 802.11) operates primarily at two layers of the OSI model:

- Layer 1 – Physical Layer (PHY): Handles the actual transmission of data over radio waves. It defines modulation schemes (OFDM, DSSS), frequency bands (2.4 GHz, 5 GHz), signal encoding, and data rates.
- Layer 2 – Data Link Layer (MAC Sublayer): Manages access to the shared wireless medium using CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance), handles MAC addressing, frame formatting, acknowledgement, and error detection.

The upper layers (Network, Transport, etc.) are handled by TCP/IP and are independent of Wi-Fi. Wi-Fi essentially provides the wireless equivalent of Ethernet at Layers 1 and 2.

Q2. Wi-Fi devices used in daily life – Wireless capability and Generation mapping

Below is a sample of commonly used Wi-Fi devices with their wireless properties and mapped Wi-Fi generations:

Device	Standard	Frequency	Max Speed	Wi-Fi Generation
Smartphone (Android/iOS)	802.11ax	2.4 / 5 GHz (Dual)	Up to 9.6 Gbps	Wi-Fi 6
Laptop	802.11ac	2.4 / 5 GHz (Dual)	Up to 1.3 Gbps	Wi-Fi 5
Smart TV	802.11n	2.4 GHz	Up to 300 Mbps	Wi-Fi 4
Wireless Router	802.11ax	2.4 / 5 / 6 GHz	Up to 9.6 Gbps	Wi-Fi 6E
IoT Smart Bulb	802.11b/g	2.4 GHz	Up to 54 Mbps	Wi-Fi 3 (Legacy)

Note: To check your own device – go to Settings → Wi-Fi → Connected Network → Properties/Details to find the actual standard, frequency, and link speed.

Q3. What is BSS and ESS?

BSS – Basic Service Set:

A BSS is the fundamental building block of a Wi-Fi network. It consists of a single Access Point (AP) and all the client devices (stations) associated with it. Each BSS is uniquely identified by a BSSID, which is the MAC address of the AP. A BSS can be of two types:

- Infrastructure BSS: Clients connect through an AP (most common in homes/offices).
- Independent BSS (IBSS) / Ad-hoc: Devices communicate directly without an AP.

ESS – Extended Service Set:

An ESS is a network formed by combining multiple BSSs connected through a common Distribution System (DS), typically a wired backbone (Ethernet). All APs in an ESS share the same SSID (network name), allowing clients to roam seamlessly from one AP to another without losing connectivity. This is widely used in:

- Corporate offices
- University campuses
- Airports and shopping malls

Q4. What are the basic functionalities of a Wi-Fi Access Point?

A Wi-Fi Access Point (AP) serves as the central hub of a wireless network. Its basic functionalities include:

- SSID Broadcasting: Continuously transmits the network name (SSID) as beacons so devices can discover the network.
- Authentication & Association: Verifies device credentials (password/certificates) and allows them to join the network.
- Wireless-to-Wired Bridging: Acts as a bridge between wireless clients and the wired LAN/internet.
- Channel & Frequency Management: Selects the appropriate channel and frequency band (2.4/5 GHz) to minimize interference.
- Security Enforcement: Implements WPA2/WPA3 encryption to secure data transmissions.
- Traffic Management: Prioritizes traffic using QoS (Quality of Service) settings.
- DHCP Relay: May assign or relay IP addresses to connected devices.
- Roaming Support: Supports seamless client handoff in ESS environments.

Q5. Difference between Bridge Mode and Repeater Mode

Feature	Bridge Mode	Repeater Mode
Primary Purpose	Connects two separate networks	Extends the range of an existing network
SSID	Different SSIDs possible	Same SSID as the main router
Wired Connection	Typically required	Not required (wireless only)
IP Routing	No NAT; same network	Relays packets wirelessly
Performance	Minimal overhead	May reduce throughput by ~50%
Use Case	Linking two office buildings wirelessly	Eliminating Wi-Fi dead zones at home
Device Type	AP in bridge mode	Range extender/Wi-Fi booster

Q6. What are the differences between 802.11a and 802.11b?

Feature	802.11a	802.11b
Release Year	1999	1999
Frequency Band	5 GHz	2.4 GHz
Maximum Data Rate	54 Mbps	11 Mbps
Modulation	OFDM	DSSS
Range (Indoor)	~35 meters	~38 meters
Range (Outdoor)	~120 meters	~140 meters
Interference	Low (less crowded 5 GHz band)	High (shares band with Bluetooth, microwaves)
Wall Penetration	Weaker	Better
Cost (at launch)	Higher	Lower
Compatibility	Not compatible with 802.11b	Not compatible with 802.11a

Q7. Modem/Hotspot configured at 2.4 GHz vs 5 GHz – Observed Differences

The following table summarizes the theoretical and typically observed differences when connecting a laptop to the same hotspot operating at 2.4 GHz vs 5 GHz:

Property	2.4 GHz Band	5 GHz Band
Frequency	2.4 GHz	5 GHz
Channel Width	20 / 40 MHz	20 / 40 / 80 / 160 MHz
Max Throughput	~150–300 Mbps (802.11n)	~300–1300 Mbps (802.11ac)
Signal Strength (Near AP)	Strong	Strong
Signal Strength (Far/Wall)	Good (better penetration)	Weaker (attenuates faster)
Interference	High (crowded band)	Low (less devices)
Latency	Slightly higher	Lower
Range	Longer (~40m indoor)	Shorter (~20m indoor)
Best Use Case	Large area coverage, IoT devices	HD streaming, gaming, video calls

Steps to capture: Connect device to 2.4 GHz → go to Wi-Fi Properties (Windows: Network Adapter → Status → Details; Mac: Option+Click Wi-Fi icon) → note RSSI, Link Speed, channel. Repeat for 5 GHz.

Q8. What is the difference between IEEE and WFA?

Aspect	IEEE (Institute of Electrical and Electronics Engineers)	WFA (Wi-Fi Alliance)
Type	International standards body	Industry consortium / trade group
Role	Defines the technical 802.11 standards	Certifies products for interoperability
Output	IEEE 802.11a/b/g/n/ac/ax specifications	Wi-Fi CERTIFIED™ logo and branding
Focus	Technical protocols and specifications	Market adoption, testing, certification

Aspect	IEEE (Institute of Electrical and Electronics Engineers)	WFA (Wi-Fi Alliance)
Membership	Open to engineers/researchers	Device manufacturers, chip vendors, ISPs
Example	Defines 802.11ax (Wi-Fi 6) specs	Certifies a router as 'Wi-Fi 6 Certified'

In short: IEEE writes the rule book; WFA ensures that products follow those rules and work together reliably.

Q9. Types of Wi-Fi Internet Connectivity Backhaul

The backhaul is the connection between the local network/AP and the internet. Common types include:

Backhaul Type	Technology	Typical Speed	Use Case
Fiber Optic (FTTH/FTTP)	Light through glass fiber	100 Mbps – 10 Gbps	Urban homes, offices
DSL (ADSL/VDSL)	Telephone copper lines	1 – 100 Mbps	Suburban/rural homes
Cable (DOCSIS)	Coaxial cable (HFC)	25 Mbps – 1 Gbps	Cable TV areas
Cellular (4G LTE / 5G)	Mobile radio network	20 – 1000 Mbps	Mobile hotspots, remote areas
Fixed Wireless Access (FWA)	Point-to-point microwave	10 – 500 Mbps	Rural/suburban last-mile
Satellite (VSAT/Starlink)	Geostationary/LEO satellites	20 – 500 Mbps	Very remote areas
Microwave/Radio Link	Licensed radio spectrum	10 – 1000 Mbps	Enterprise/ISP backhaul

Note: Check your home/college internet connection type by asking your ISP or checking the router's WAN settings. For VIT Chennai, the campus likely uses Fiber Optic backhaul with a dedicated leased line.

Q10. Wi-Fi Topologies and Their Use Cases

Topology	Description	Use Case
Infrastructure (BSS)	Clients connect through a central Access Point (AP). Most common setup.	Home Wi-Fi, office networks, cafes
Ad-hoc / IBSS	Devices connect directly to each other without any AP. Peer-to-peer.	Laptop-to-laptop file sharing, gaming
Extended Service Set (ESS)	Multiple APs connected via wired backbone sharing the same SSID for seamless roaming.	University campuses, airports, malls
Mesh Network	Multiple nodes communicate with each other wirelessly. Self-healing, no single point of failure.	Smart homes, city-wide Wi-Fi, warehouses
Point-to-Point (PtP)	Two directional antennas link two locations wirelessly over long distances.	Building-to-building connectivity
Point-to-Multipoint (PtMP)	One central AP broadcasts to multiple remote nodes simultaneously.	ISP wireless distribution, rural connectivity
Wi-Fi Direct	Devices connect directly using Wi-Fi without a router (P2P but with Wi-Fi speeds).	Screen mirroring, printer connection, file transfer